

SAMSUNG

Submittal Data

AE055FCYDCG/AA EHS Air-to-Water Heat Pump, Monobloc

Job Name: _____ Location: _____
Purchaser: _____ Engineer: _____
Submitted to: _____ Reference Approval Construction
Unit Designation: _____ Schedule #: _____

General Information

- The system shall be capable of heating and cooling operation, with heating operation down to -22°F (-30°C).
- The outdoor unit shall have a base pan heater as standard for optimal defrost cycle water drainage.
- The outdoor unit connects to one of two indoor unit options; a hydro unit, or the ClimateHub, which features an integral domestic hot water tank.
- Auto-restart after power loss
- The compressor shall be soft-start, inverter-driven type to minimize inrush current

Construction

- Outdoor unit: Galvanized steel with a baked-on powder-coated finish for durability.
- The outdoor unit shall feature a brazed-plate refrigerant-to-water heat exchanger with a built-in heater for freeze protection.
- A refrigerant-cooled inverter circuit facilitates heating operation up to 109°F (43°C).
- The outdoor unit coil shall feature a 3000-hour salt spray coating as standard.
- The compressor is mounted using spring grommets to reduce vibration.
- The compressor is blanketed with a two-layer grooved felt to reduce sound transmission.
- The fan features a multi-serration edge to minimize turbulence, increase efficiency, and reduce noise levels.
- The fan motor features a 10-pole rotor to reduce vibration
- The base pan features a sloped design to better facilitate water drainage during defrost.

Refrigerant System

- The system shall utilize low-GWP R-32 refrigerant.
- Refrigerant flow shall be controlled by an electronic expansion valve.

Controls

- The outdoor unit shall communicate to the indoor unit via 16-18 AWG X 2 stranded cable.
- The outdoor unit shall be controlled using the AI Home interface included with the indoor unit.
- The system shall support a cooling setpoint, a space heating setpoint, and a domestic hot water setpoint.

Convenience

- The outdoor unit features three optional levels of silent-mode operation.
- A snow-accumulation-prevention function reduces snow build up when the outdoor unit fan is idle for extended periods of time.
- The system can be configured as heat pump (heating & cooling) or heating only via outdoor unit option setting.
- Maximum Current Control configurable from 50% - 100%.

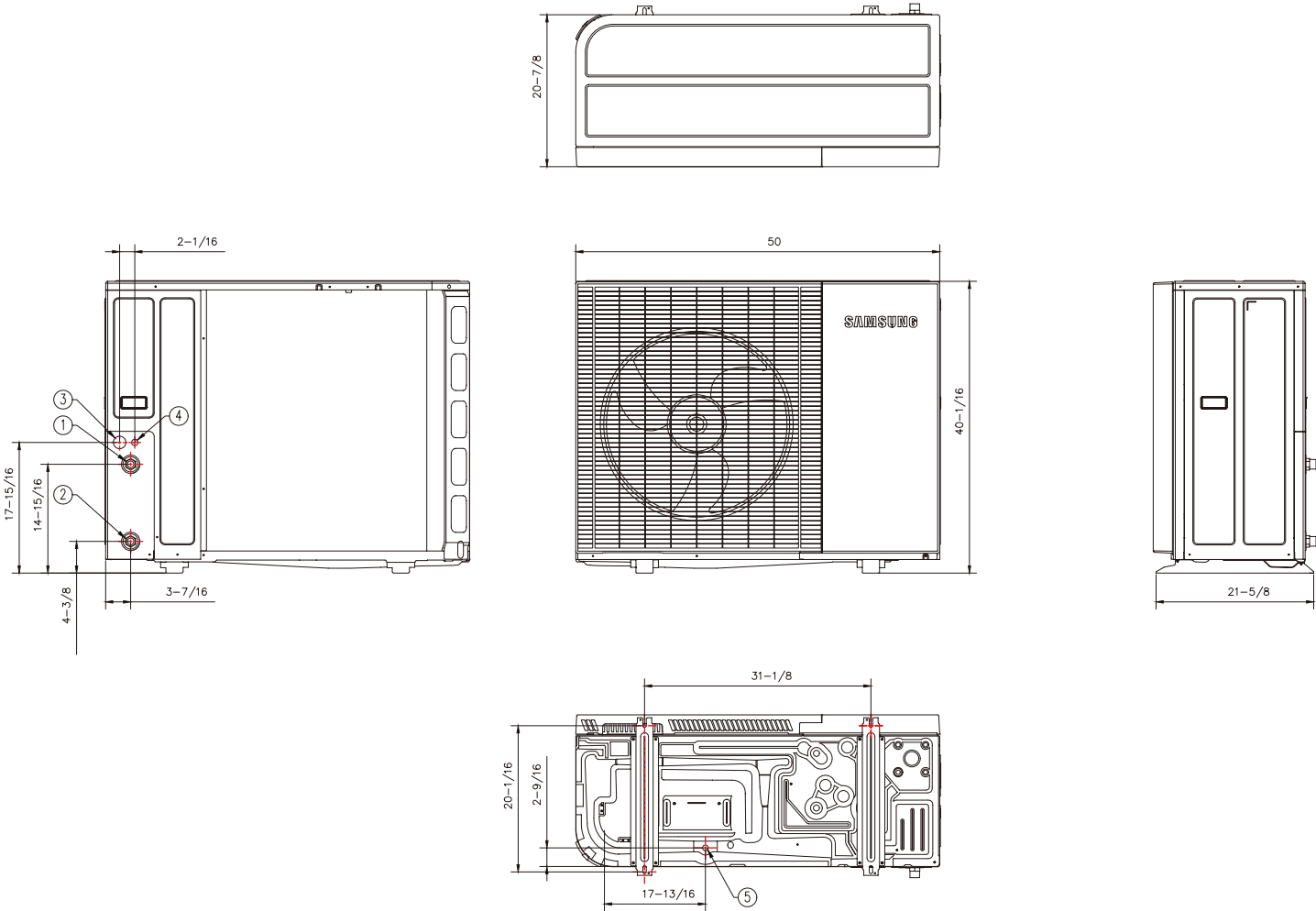


Specifications

Model	Outdoor Unit Model Number		AE055FCYDCG/AA
Performance	Nominal Capacity	Cooling / Heating (Btu/h)	54,600 / 54,600
	System Cooling Performance ¹	EER A95W65 (Cond #1)	4.10
		EER A95W45 (Cond #2)	2.95
	System Heating Performance ¹	COP A45W95 (Cond #1)	4.75
		COP A45W113 (Cond #2)	3.55
		COP A45W131 (Cond #3)	3.35
		COP A35W95 (Cond #4)	3.80
Power	Voltage	ø / V / Hz	1 / 208-230V / 60 Hz
	Working Voltage Range (VAC)		198 - 264
	Operating Current (Cond. #1 / Cond. #2))	Cooling (A)	19.28 / 22.61
		Heating (A)	16.64 / 21.3
	Max. Breaker	Amps	40.0
	Min. Circuit Ampacity	Amps	32.0
Dimensions (Net)	W X H X D	in.	50 x 40.08 x 20.87
	Weight	lbs.	302.00
Sound Level	Sound Power	dB(A)	60
	Sound Pressure	dB(A)	47
Operating Temperatures	Outdoor Ambient	Cooling	50 - 115°F (10 - 46°C)
		Heating	-22 - 109°F (-30 - 43°C)
	Leaving Water Temperature	Cooling	41 - 77°F (5 - 25°C)
		Heating	59 - 158°F (15 - 70°C)
		Domestic Hot Water	59 - 158°F (15 - 70°C)
Pipe Connections	Type		NPT male
	Diameter	in.	1"
Refrigerant	Type		R-32
	Factory Charge	lbs.	7.28
	Heat Exchanger Type		Copper & Stainless Steel Brazed Plate
Compressor	Type		Inverter-Driven Scroll
	RLA	Amps	18.9
Water Data	Water Flow Rate	Min / Nominal / Max (GPM)	1.8 / 10.7 / 15.3
	Max Water Pressure	PSI	43.5
	Strainer		50-mesh
Condenser Fan	Type		Propeller
	Watts / CFM (max.)		122 / 3,355
Corrosion Prevention	ISO-9227 - The condenser heat exchanger (air) showed no unusual rust or corrosion development to 3,000 hours.		
Safety	Certifications	UL 60335-2-40	
	Devices	PCB fuses, terminal block thermal fuse, current transformer, over-voltage protection, crankcase heating, PHE heater, temperature limit protection logic, compressor overload sensing	

Dimensional Drawing AE055FCYDCG/AA

Unit: Inches



No.	Name	Description
1	Water pipe (out)	NPT male, 1"
2	Water pipe (in)	NPT male, 1"
3	Power wiring conduit	Φ44
4	Communication wiring conduit	Φ22
5	Drain holes	Connect with the provided drain plug